



EMS-Flow

AVFS Area Velocity Flow Sensor — Configuration & Monitoring Software

User Manual

Software version 1.28.1 · June 2026

Environmental Data Services

www.e-d-s.com.au · 1300 721 683

1. Introduction

EMS-Flow is a Windows desktop application for commissioning and monitoring AVFS area velocity flow sensors - Dual 1 MHz Doppler Ultrasonic Velocity Instruments - over an RS-485 serial connection using the Modbus RTU protocol. Field technicians use it to set up a sensor on site (pipe or channel profile, level, velocity calibration and logging behaviour) and to view live measurements at the asset.

Reliability is built in: every setting EMS-Flow writes is confirmed by reading the value back from the sensor, every value it displays was read from the device moments before, and the communication layer rejects stale or corrupted serial frames rather than acting on them. If a value cannot be confirmed, EMS-Flow says so in the log rather than guessing.

In a hurry? The separate EMS-Flow Quick Start Guide covers connect-and-commission on a single page. This manual is the full reference for every screen and setting.

2. What You Need

- A Windows 10/11 (64-bit) or macOS 11+ laptop. Windows-on-ARM (e.g. Parallels on Apple Silicon) is supported by the standard installer.
- A USB-to-RS-485 adapter (CH340, CP2102 or FTDI chipsets are all suitable).
- An AVFS sensor wired for power (8–14 V DC) and RS-485.

2.1 Wiring

Wire colour	Function	Terminal
Red	Power +	8–14 V DC
Black	Power –	GND
Blue	RS-485 A	485-A on adapter
Green	RS-485 B	485-B on adapter

Factory communication defaults: 9600 baud, 8 data bits, no parity, 1 stop bit, slave address 1.

2.2 Adapter Drivers

Most adapters need a one-time driver install before Windows shows them as a COM port. Identify the chipset (printed on the adapter or its listing), install the matching driver, then unplug and replug the adapter.

Chipset	Driver download
FTDI / FT232	ftdichip.com/drivers/vcp-drivers/

3. Installing & Updating

1. Close EMS-Flow if an older version is running.
2. Double-click EMS-Flow-Setup-<version>.exe and follow the prompts. The installer upgrades an existing copy in place — you never need to uninstall first.
3. Launch from the EMS-Flow desktop shortcut or Start Menu.

Windows SmartScreen. On a first install Windows may show an "unrecognised app" notice. Click "More info" then "Run anyway". Automatic updates after installation are not affected.

3.1 Automatic Updates

EMS-Flow checks for updates each time it starts (an internet connection is needed for the check; working offline is fine and produces no errors). A new version downloads silently in the background, then a banner offers to install it.

- **Restart now** applies the update immediately (disabled while connected to a sensor — disconnect first).
- **Later** dismisses the banner; the update installs automatically next time you close the app.

One copy at a time. EMS-Flow runs as a single instance. Launching it again focuses the existing window rather than opening a second copy, so two copies can never fight over the COM port.

4. Connecting to a Sensor

EMS-Flow is designed to connect on its own. Plug in the RS-485 adapter (Blue → 485-A, Green → 485-B) and power the sensor — when a single adapter is present, the app connects automatically, the status dot turns green, and the sensor's saved configuration loads into the Site Setup screen.

4.1 If It Doesn't Connect Automatically

1. Open the Port dropdown in the top bar and select your adapter (a bullet • marks likely RS-485 adapters).
2. Check Baud (default 9600) and Slave address (default 1) match the sensor.
3. Press **Connect**.

4.2 Automatic Behaviours

- **Live detection** — the port list updates by itself as adapters are plugged in or removed; no need to press refresh.
- **Auto-reconnect** — if the adapter is unplugged or the sensor loses power, EMS-Flow reconnects automatically when it returns.
- **Deliberate disconnect** — pressing Disconnect is respected; the app will not auto-reconnect until you reconnect or restart.

Important. Only one program can hold a COM port at a time. If another copy of EMS-Flow, the ADV logger tool, or other serial software has the port open, connection fails with "Access denied" and the log tells you what to close.

5. Setup Wizard — Commissioning a Site

The Setup Wizard (first item in the sidebar) takes a powered, connected sensor through a complete commissioning in five steps. Use the Back and Next buttons to move between them.

The screenshot displays the EMS-Flow AVFS Setup Wizard interface. At the top, there's a header with the EDS logo, company name, and version information. Below this is a sensor status bar showing 'Searching for sensor...' and buttons for 'CONNECT', 'TAKE READING', 'READ CONFIG', and 'CONNECTION SETTINGS'. The left sidebar lists navigation options: 'GET STARTED' (Setup Wizard), 'SENSOR' (Dashboard, Site Setup, Output to Logger, Flow Totals), and 'ADVANCED' (Data Registers, Diagnostics, Device Info). The main content area shows a progress bar with five steps: 1. Zero depth, 2. Site details, 3. Averaging, 4. Verify, and 5. Commission. Step 1 is currently active. The instructions for Step 1 state: 'Before the sensor goes into the flow, zero its depth reading in open air so it starts from an accurate datum. Do this first, while the sensor is dry and in your hand.' A callout box advises: 'Hold the sensor in open air with nothing on the sensor face before zeroing. Zeroing while wet or submerged will offset every future depth and flow reading.' A button labeled 'ZERO DEPTH (SENSOR IN AIR)' is present. At the bottom, there are 'BACK' and 'NEXT' buttons.

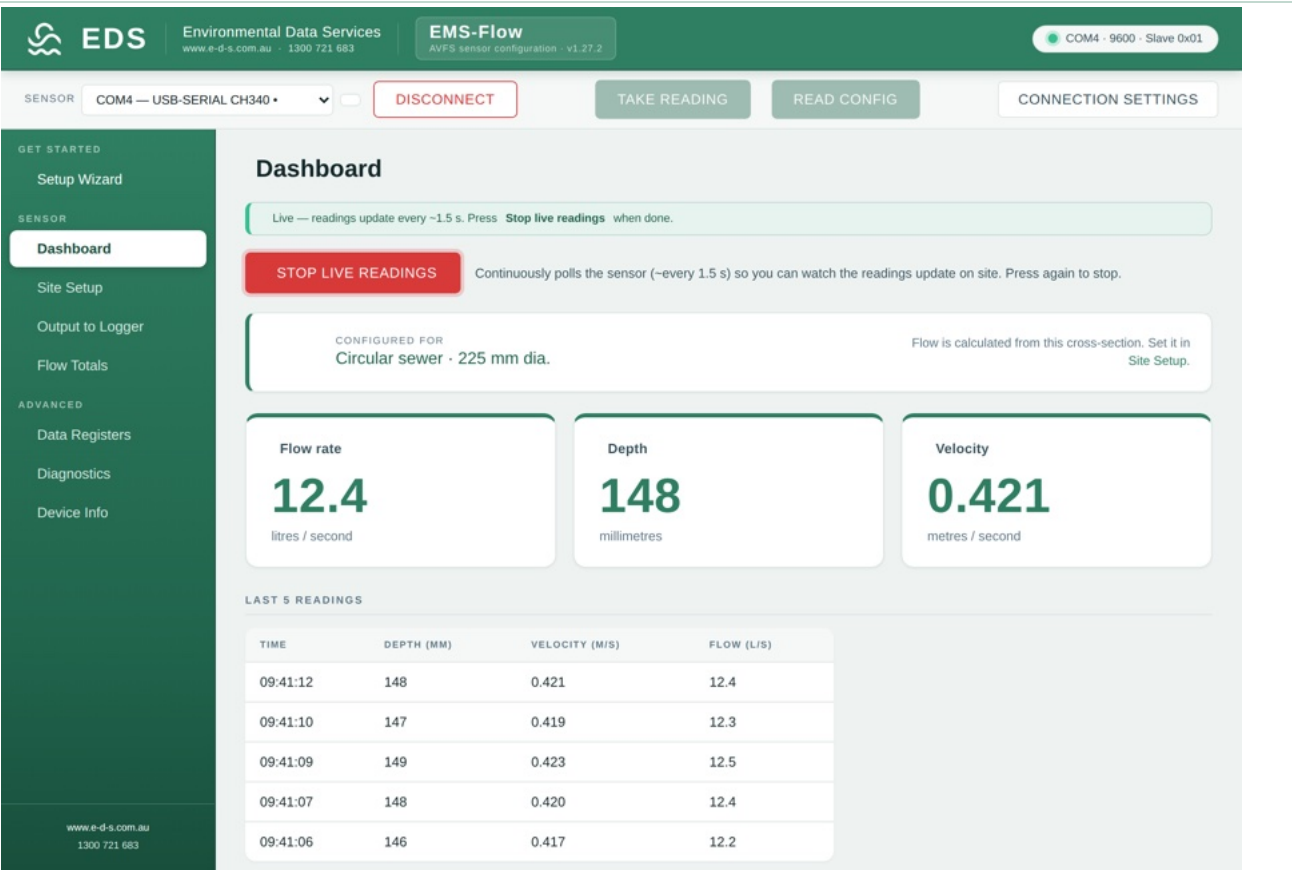
The Setup Wizard, Step 1 — the five-step progress bar runs across the top.

1. **Step 1 — Zero the depth.** With the sensor held in air, dry, press "Zero depth (sensor in air)". This sets the dry datum that depth is measured from. The sensor must be out of the water for this step.
2. **Step 2 — Describe the site.** Choose the shape (rectangular, trapezoidal, circular, egg-shaped or U-shaped) and enter its dimensions in millimetres. For circular pipe, enter the nominal bore, e.g. 150, 225, 300.
3. **Step 3 — Set the averaging time.** Choose how quickly the smoothed velocity responds. 5 seconds suits most sites; use longer for surcharge or tidal flow.
4. **Step 4 — Verify the readings.** Confirm the live values look sensible before committing.
5. **Step 5 — Commission the sensor.** Press "Save & start sensor". Every setting is written, read back from the sensor to verify it stored correctly, then a live reading confirms the sensor is measuring. Any value that fails verification is flagged.

Verified writes. A tick (✓) in the log means the value was written and read back identical. A warning means the sensor acknowledged but stored something different — write it once more; if it sticks the second time it was a commit-timing quirk in the meter.

6. Dashboard — Live Readings

The Dashboard shows the sensor's live measurements. Press **Live readings** to poll automatically (about every 1.5 s); press again to stop. The banner is green when connected and ready, and changes to confirm when live. Live polling pauses automatically while configuration is read or written, then resumes.



The Dashboard during live readings, showing flow, depth and velocity with the rolling last-5 table.

Reading	Description
Flow rate	Instantaneous flow in litres/second, from velocity × wetted area
Depth	Water depth above the zeroed datum, in millimetres
Velocity	Smoothed flow velocity in m/s
Secondary cards	Instant velocity, wetted area, temperature, signal strength, conductivity
Last 5 readings	A rolling table beneath the cards, newest at the top

The **Configured for** bar shows the pipe/channel the flow is being calculated from, so the measurement context is always visible.

Battery-saving sites. In interval mode the velocity shown is the average over the configured window, refreshed at the end of each window — not a five-times-a-second live value. This is expected, not a fault.

7. Site Setup

Site Setup holds every sensor configuration card. Each card writes when its own Write button is pressed (or by pressing Enter in its value field). Fields show values loaded from the sensor on connection. The Read Config button at the top re-reads everything from the sensor.

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EMS-Flow
AVFS sensor configuration - v1.27.2

COM4 - 9600 - Slave 0x01

SENSOR COM4 - USB-SERIAL CH340 • DISCONNECT TAKE READING READ CONFIG CONNECTION SETTINGS

GET STARTED
Setup Wizard

SENSOR
Dashboard
Site Setup
Output to Logger
Flow Totals

ADVANCED
Data Registers
Diagnostics
Device Info

Site Setup

Set the sensor up to match the sewer or channel it is installed in. All depths and dimensions are entered in millimetres (mm).

FIRST, ON SITE: ZERO THE DEPTH DATUM

ZERO LEVEL — SET THE DRY DATUM
Do this first when you arrive on site, before configuring anything else. It re-zeros the depth reading with the sensor out of the water, in open air, so dry reads as zero. (Different from Depth calibration below, which matches the reading to a live dip.)

Only zero the level with the sensor held in air. Zeroing while submerged will offset every future depth and flow reading.

ZERO LEVEL (SENSOR IN AIR)

VELOCITY AVERAGING

VELOCITY AVERAGING TIME (SENSOR READING TIME)
15 seconds — steadier reading

How long the sensor averages its velocity readings before reporting a value. A longer time gives a steadier reading in turbulent flow but responds more slowly to real changes.

WRITE AVERAGING TIME

SEWER / CHANNEL PROFILE

SHAPE

Site Setup — each card configures one aspect of the sensor and writes independently.

7.1 Zero Level — Set the Dry Datum

1. Hold the sensor in air, completely out of the water.
2. Press "Zero level (sensor in air)". The current reading becomes the zero point for all depths.

7.2 Velocity Averaging Time

Controls how quickly the smoothed velocity responds. Choose the option for the site, then press "Write averaging time".

Option	Use for
1 second	Fastest, live response — diagnostics and bench work
5 seconds	Recommended for most sites
15 seconds	Steadier reading on noisy flow
60 seconds	Very steady — surcharge or tidal sites

7.3 Sewer / Channel Profile

1. Choose the Shape. The dimension fields change to suit it.
2. Enter the dimensions in millimetres.

3. Press "Write sewer profile". Each dimension is written, read back and verified; the log shows a ✓ per value.

Circular pipes and standard bores. The sensor stores pipe size as a radius rounded down to the nearest centimetre, so an exact read-back of a 150 mm pipe returns ~140 mm. EMS-Flow handles this automatically: diameters within 30 mm of a standard sewer bore (100, 150, 200, 225, 250, 300, 375, 450, 525, 600, 675, 750, 825, 900, 1050, 1200, 1350, 1500) are displayed and verified as that standard size. Non-standard diameters are shown exactly as reported.

7.4 Level Adjustment — Match a Manual Dip

Use on site when the sensor's depth disagrees with a manual measurement.

1. Press "Read depth" to fetch what the sensor currently reads.
2. Enter the true depth from your manual measurement.
3. Press "Adjust depth to match" (or Enter). EMS-Flow computes and writes the offset, then verifies it.

7.5 Velocity Correction

Applies a proportional correction so the sensor's velocity matches a reference measurement.

1. Press "Read velocity" to capture the current reading.
2. Enter the reference velocity from your independent measurement.
3. Press "Apply velocity correction" (or Enter). The factor is calculated, written and verified.

7.6 Sampling / Logging Mode

Continuous keeps the sensor measuring continuously (~5 times/second, ~55 mA at 12 V). **Interval averaging** is the battery-saving mode: the sensor wakes, averages flow over the window, logs it and sleeps (~17 mA average).

1. Choose the Sampling mode.
2. For interval mode, set the Averaging interval — between 1 and 60 seconds.
3. Press "Write logging mode". The setting is written and read back to confirm what the sensor stored.

60-second maximum. The current sensor firmware cannot store windows longer than 60 seconds; EMS-Flow enforces this so the value you set is always the value the sensor uses.

8. Output to Logger — Communication Settings

Sets the Modbus identity the sensor presents to a data logger or telemetry unit: slave address (1–254) and connection speed (2400–115200 baud).

The screenshot shows the EMS-Flow web interface. At the top, there's a header with the EDS logo, company name, and a status bar showing 'COM4 - 9600 - Slave 0x01'. Below the header, a navigation bar includes a 'SENSOR' dropdown, a 'DISCONNECT' button, and 'TAKE READING', 'READ CONFIG', and 'CONNECTION SETTINGS' buttons. The left sidebar lists navigation options: 'GET STARTED' (Setup Wizard), 'SENSOR' (Dashboard, Site Setup, Output to Logger, Flow Totals), and 'ADVANCED' (Data Registers, Diagnostics, Device Info). The 'Output to Logger' option is selected. The main content area is titled 'Output to Logger / RTU'. It contains a text box explaining that the AVFS reports readings to a data logger or RTU over a two-wire RS-485 line. Below this is a 'WIRING TO THE LOGGER / RTU' section with a legend: Red for Power + (8–14 V DC), Black for Power – (0 V / ground), Blue for RS-485 A (D+), and Green for RS-485 B (D–). A note states to keep the RS-485 run under 200 m and use shielded twisted pair. The 'LOGGER SETTINGS' section has two input fields: 'SENSOR ADDRESS (1-254)' with the value '1' and 'CONNECTION SPEED (BAUD)' with a dropdown set to '9600'. A note below these fields states that each sensor needs its own address and that the speed must match the logger's. An orange button 'SAVE LOGGER SETTINGS TO SENSOR' is at the bottom. A warning message at the bottom of the settings section says: 'Changing the address or speed disconnects this app from the sensor. Reconnect using the new settings afterwards.'

Output to Logger — set the address and baud the telemetry unit expects.

1. Enter the address and/or baud the logger expects.
2. Press "Save logger settings to sensor".
3. Reconnect using the new settings — change Baud and Slave in the top bar to match, then Connect.

Take care. These settings take effect on the sensor immediately. If you lose track of them, the Diagnostics page can help, or power-cycle and try the factory defaults (9600, address 1).

9. Flow Totals

Shows the accumulated flow totaliser. **Write totaliser** presets the total to the value entered — use 0 to reset. Useful when replacing a sensor so the new unit continues from the old total.

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EMS-Flow
AVFS sensor configuration - v1.27.2

COM4 - 9600 - Slave 0x01

SENSOR COM4 — USB-SERIAL CH340 •

DISCONNECT TAKE READING READ CONFIG CONNECTION SETTINGS

GET STARTED
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Flow totals

Cumulative volume passing the sensor. Forward is normal downstream flow; reverse indicates backflow or surcharge.

FORWARD VOLUME (DOWNSTREAM) 0 m³

REVERSE VOLUME (BACKFLOW) 0 m³

NET VOLUME 1284.730 m³

RESET / PRESET NET FLOW TOTALISER

Use this to zero the totaliser at commissioning, or preset it to match an existing meter reading. Enter 0 to reset.

0 m³ WRITE TOTALISER

Flow Totals — the accumulated net total, with a preset/reset control.

Confirmation required. Writing the totaliser permanently overwrites the accumulated total on the sensor, so EMS-Flow asks you to confirm first. Record the old value beforehand if it matters.

10. Data Registers

An advanced live view of every sensor register EMS-Flow knows — name, address, current value and access. Useful for spot-checking a specific register or watching a value the Dashboard does not show. Values refresh while live readings are on.

11. Diagnostics

Advanced tools for troubleshooting the RS-485 link — needed only if a sensor is not responding.

The screenshot shows the EMS-Flow Diagnostics interface. The top header includes the EDS logo, company name, and the EMS-Flow version. Below this, a sensor selection dropdown is set to 'COM4 - USB-SERIAL CH340', accompanied by a 'DISCONNECT' button and buttons for 'TAKE READING', 'READ CONFIG', and 'CONNECTION SETTINGS'. A left sidebar provides navigation options under 'GET STARTED' (Setup Wizard), 'SENSOR' (Dashboard, Site Setup, Output to Logger, Flow Totals), and 'ADVANCED' (Data Registers, Diagnostics, Device Info). The main 'Diagnostics' section contains a description of its purpose and three functional areas: a 'FRAME BUILDER — FC03 READ REQUEST' with input fields for slave address (01), start register (0000), and quantity of registers (0002); a 'COMMS LOG' section with a 'CLEAR' button and a large dark area for log entries; and 'DEFAULT FRAME SETTINGS' showing the protocol as 'Modbus RTU, master-slave half-duplex' and the frame configuration as '1 start bit, 8 data bits, 1 stop bit, no parity'.

Diagnostics — the Modbus frame builder and the live communication log.

11.1 Frame Builder

Builds a raw Modbus FC03 read request byte by byte, with the CRC calculated for you:

1. Enter the slave address (hex), start register and quantity of registers.
2. The frame bytes are shown live, including the CRC.
3. Send it and watch the reply in the comms log, or copy the bytes into another tool.

11.2 Comms Log

Every frame EMS-Flow sends and receives is logged with a timestamp, in full hex, on every page. This is the first thing to capture when reporting a sensor problem — it shows exactly what was asked and exactly what the sensor answered. Clear empties the log.

12. Troubleshooting

Symptom	What to do
No adapter in the dropdown	Install the adapter driver (2.2), replug, press refresh. On Parallels confirm the adapter is routed to Windows. Check Device Manager → Ports (COM & LPT) for a warning triangle.
"Access denied" when connecting	Another program holds the port. Close other EMS-Flow windows (Task Manager → Details → EMS-Flow.exe), the ADV logger tool, or terminal software, then Connect.
"Timeout — no response from sensor"	Check baud and slave address match the sensor. Swap the A/B wires. Confirm the sensor has power (8–14 V DC).
Readings all zero	Wait ~5 s after powering the sensor. Confirm water is at least 20 cm above the sensor.
"no clean reply (field left unchanged)"	The sensor answered inconsistently for that register. Press Read Config again; if it persists, check wiring quality and cable length (485 max 200 m, shielded twisted pair).
A written value reads back different	Write it once more. A repeat mismatch is logged with both values — capture the comms log and report it.
Desktop shortcut shows a generic icon	Windows caches icons. Run ie4unit.exe -show from Command Prompt, or restart the PC.

13. Reference

13.1 Communication Defaults

Setting	Factory default
Protocol	Modbus RTU, master–slave, half duplex
Serial format	9600 baud, 8 data bits, no parity, 1 stop bit
Slave address	1

13.2 Sensor Measurement Specifications

Parameter	Specification
Measurement principle	Dual 1 MHz Doppler ultrasonic velocity
Velocity range	0–10 m/s (± 10 m/s with direction identification)
Velocity accuracy	1% $\pm$ 0.005 m/s (resolution 0.001 m/s)
Water level range	0–10 m (accuracy 0.2% FS + 0.001 m)
Temperature	0–60 °C (± 1 °C)
Conductivity	0–10 mS/cm (2%)
Power	8–14 V DC · 55 mA continuous / 17 mA interval mode (at 12 V)

13.3 Document

Item	Detail
Applies to	EMS-Flow v1.28.1
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